Linear Functions & Inequalities

ANSWER KEY

Slope and slope-intercept form

1 Find the slope of the line passing through $(2, 3)$ and $(6, 11)$.
Slope $= \frac{11-3}{6-2} = \frac{8}{4} = 2$.

2 Write the equation of the line with slope -3 passing through $(1, 4)$, in slope-intercept form.
$y - 4 = -3(x-1)$, so $y = -3x + 3 + 4 = -3x + 7$.

Parallel and perpendicular lines

3 Find the equation of the line parallel to $y = 2x - 5$ passing through $(3, 4)$.
Parallel lines share the same slope, $m = 2$. $y - 4 = 2(x-3)$, so $y = 2x - 6 + 4 = 2x - 2$.

4 Find the equation of the line perpendicular to $y = \frac{1}{2}x + 3$ passing through $(4, 1)$.
The perpendicular slope is the negative reciprocal: $m = -2$. $y - 1 = -2(x-4)$, so
$y = -2x + 8 + 1 = -2x + 9$.

Finding equations from two points or a table

5 A linear function passes through $(0, 5)$ and $(4, 17)$. Find its equation.
Slope $= \frac{17-5}{4-0} = \frac{12}{4} = 3$. Since $(0, 5)$ is the y-intercept, the equation is $y = 3x + 5$.

6 A table shows $x = 1, y = 7$ and $x = 3, y = 15$ for a linear function. Find the equation.
Slope $= \frac{15-7}{3-1} = \frac{8}{2} = 4$. Using point $(1, 7)$: $y - 7 = 4(x-1)$, so $y = 4x - 4 + 7 = 4x + 3$.

Solving and graphing linear inequalities

7 Solve the inequality $3x - 4 < 11$, and graph the solution on a number line description.
$3x < 15$, so $x < 5$. On a number line, this is an open circle at 5 with shading extending to the left (all values less than 5).

8 Solve the inequality $-2x + 7 \geq 1$, being careful with the direction of the inequality sign.
$-2x \geq -6$. Dividing by -2 flips the inequality: $x \leq 3$.

Systems of linear inequalities

9 A system requires $y > 2x - 1$ and $y \leq -x + 5$. Determine whether the point $(1, 2)$ satisfies both inequalities.

Check the first: $2 > 2(1) - 1 = 1$? True. Check the second: $2 \leq -1 + 5 = 4$? True. Since $(1, 2)$ satisfies both inequalities, it lies in the solution region.

Interpreting linear inequalities in context

10 A student has 50 dollars and wants to buy notebooks costing 6 dollars each, keeping at least 8 dollars in reserve. Write an inequality for the number of notebooks n the student can buy, and solve it.

The amount spent, $6n$, must leave at least 8 dollars: $50 - 6n \geq 8$, so $-6n \geq -42$, giving $n \leq 7$ (dividing by a negative number flips the inequality). The student can buy at most 7 notebooks.

11 A factory produces at least 200 units per day but at most 350 units per day. Write a compound inequality for the daily production p, and state whether $p = 180$ is a valid production level.

$200 \leq p \leq 350$. Since $180 < 200$, the value $p = 180$ does NOT satisfy the inequality — it is below the minimum required production level.

Comparing rates of change between two linear functions

12 Function f is given by $f(x) = 3x + 2$. Function g is defined by the table: $x = 0, g = 1$ and $x = 2, g = 9$. Which function has the greater rate of change?

f has a rate of change (slope) of 3. For g: slope $= \frac{9-1}{2-0} = \frac{8}{2} = 4$. Since $4 > 3$, function g has the greater rate of change.

Finding the intersection of two lines graphically and algebraically

13 Two lines are given by $y = 2x - 1$ and $y = -x + 8$. Find their point of intersection, and explain what this point represents graphically.

Set equal: $2x - 1 = -x + 8$, so $3x = 9$, giving $x = 3$. Then $y = 2(3) - 1 = 5$. The lines intersect at $(3, 5)$, which represents the single point where both lines cross on a graph — the unique solution to the system formed by the two equations.

Absolute value equations and inequalities

14 Solve $|2x - 3| = 9$.

Either $2x - 3 = 9$, giving $x = 6$, or $2x - 3 = -9$, giving $x = -3$. Solutions: $x = 6$ or $x = -3$.

15 Solve $|x + 4| < 6$, giving your answer as an interval.

$-6 < x + 4 < 6$, so $-10 < x < 2$.

Writing a linear inequality from a verbal description

16 A shipping company charges 3 dollars per pound but offers free shipping if the package weighs less than 2 pounds. Write a piecewise description of the cost $C(w)$ as a function of weight w, and find the cost of a 5-pound package.

$C(w) = 0$ for $w < 2$ and $C(w) = 3w$ for $w \geq 2$. For a 5-pound package: $C(5) = 3(5) = 15$ dollars.

17 A parking garage charges 5 dollars for the first hour and 2 dollars for each additional hour. Write an equation for the total cost C after h hours (for $h \geq 1$), and find the cost for 6 hours.

$C(h) = 5 + 2(h - 1)$ for $h \geq 1$ (since only the hours after the first are charged at the additional rate). For 6 hours: $C(6) = 5 + 2(5) = 5 + 10 = 15$ dollars.

Graphing a system of inequalities and identifying the feasible region

18 A system requires $x + y \leq 10$ and $x \geq 2$ and $y \geq 0$. Determine which of the points $(3, 4)$ and $(8, 5)$ lie in the feasible (solution) region, checking each condition.

For $(3, 4)$: $3 + 4 = 7 \leq 10$ ✓, $3 \geq 2$ ✓, $4 \geq 0$ ✓ — all conditions satisfied, so $(3, 4)$ is in the feasible region. For $(8, 5)$: $8 + 5 = 13 \leq 10$? False — this point fails the first condition, so $(8, 5)$ is NOT in the feasible region.

Rate of change from a real-world linear model

19 A water tank starts with 500 gallons and drains at a constant rate. After 4 hours, it contains 340 gallons. Write a linear equation for the volume V after t hours, and predict when the tank will be empty.

Rate of change: $\frac{340 - 500}{4 - 0} = \frac{-160}{4} = -40$ gallons per hour. Equation: $V = 500 - 40t$. The tank is empty when $V = 0$: $500 - 40t = 0$, so $t = 12.5$ hours.